

ZIQI(OLIVIA) WANG

✉ ziqi.wang0908@rutgers.edu | 📞 +1-732-322-3515 | 🌐 https://ziqiwan0908.github.io/

EDUCATION

- **Rutgers University, New Brunswick, NJ, United States** 01/2025 - 12/2026(*expected*)
Master of Science in Computer Science, Department of Computer Science **GPA: 4.0/4.0**
- **Hebei University of Technology, Tianjin, China** 09/2020 - 06/2024
Bachelor of Literature in International Education of Chinese, School of Humanities and Law **GPA: 3.98/4.0**
- **2nd Prize in Graduation Scholarship**, 2023-2024 Academic Year, Hebei University of Technology 06/2024

PUBLICATION

- **AURORA: Attention-intended Dual Verification with Adaptive Rollback for LLM Agents** (project page)
Ziqi Wang, Huaiyuan Yao, Hua Wei, Hang Zhou, Xiao Luo Under Review at IROS 2026
- **RAGRouter-Bench: A Dataset and Benchmark for Adaptive RAG Routing** (arXiv:2602.00296)
Ziqi Wang, Xi Zhu, Shuhang Lin, Haochen Xue, Minghao Guo, Yongfeng Zhang Under Review at ARR 2026

RESEARCH EXPERIENCE

- **AURORA: LLM Agent Trajectory Verification** 11/2025 - 03/2026
Research Assistant, Supervisor: Xiao Luo, University of Wisconsin-Madison
 - **Dual-Head Verifier Design:** Designed an attention-based dual-head verifier using distinct masking strategies to produce local (step validity) and global (trajectory coherence) scores within a single shared encoder.
 - **Context Isolation Mechanism:** Developed AURORA-Isolation to mask trajectory history from the local head before attention, preventing representation interference and resolving score convergence under shared parameters.
 - **Adaptive Rollback & Evaluation:** Implemented adaptive rollback triggered by dual-dimensional verification signals, achieving +19.2pp success rate improvement over GA-Rollback baseline on ALFWorld across 3B–14B models.
- **RAGRouter-Bench: Adaptive RAG Routing Research** 05/2025 - 01/2026
Research Assistant, Supervisor: Yongfeng Zhang, Rutgers University
 - **Benchmark Construction:** Built RAGRouter-Bench, the first adaptive RAG routing benchmark covering 7,727 queries and 21,460 documents, standardizing 5 representative RAG paradigms under a unified retriever abstraction.
 - **Dual-View Routing Indicators:** Introduced dual-view query-corpus compatibility metrics (semantic + structural) to characterize RAG paradigm applicability across factual, reasoning, and summarization tasks.
 - **Evaluation Framework & Insights:** Designed a unified effectiveness-efficiency evaluation protocol, revealing that no single RAG paradigm universally dominates and corpus properties critically shape routing decisions.

WORKING EXPERIENCE

- **Siemens Ltd., China** 05/2025 - 08/2025
LLM and Data Analysis Intern, Digital Industrial Software Dept. Beijing, China
 - **Fine-tuned Industrial LLM:** Fine-tuned an industrial-specific Qianwen2.5 32B model by combining LoRA and prompt engineering, reducing hallucinations and enhancing accuracy on technical terminology by 12%.
 - **Built RAG Knowledgebase:** Cleaned and structured extracted multimodal data with Pandas to build an industrial knowledge base in Elasticsearch with over 2,000 entities, providing the dataset for a downstream RAG system.
- **Turingops Co., Ltd.** 04/2024 - 11/2024
AI Engineer Intern, Product Development Dept. Shanghai, China
 - **Built Intent Classifier:** Co-developed a core user intent recognition module, using prompt engineering, embedding similarity, a fine-tuned classifier, and an LLM reflection mechanism to achieve 85% stable accuracy.
 - **Optimized Query Robustness:** Contributed to a query preprocessing module by using sub-question generation, query rewriting and structural decomposition to boost average answer accuracy on complex queries by 16%.
- **SOHU.com Limited** 01/2024 - 04/2024
Back-end Development Intern, Product Development Dept. Wuhan, China
 - **Back-end Developed:** Contributed to building microservices and RESTful APIs (Spring Boot) and optimizing database persistence layers, including implementing CRUD operations with JPA.
- **Danfoss Ltd., China** 03/2023 - 12/2023
Strategic Intern, Security Operation Center, Danfoss Group IT Tianjin, China
 - **Cybersecurity Maintenance and Research:** Developed automated Shell scripts for network traffic analysis and server diagnostics, supporting monitoring and failure analysis for a 1000+ device cluster and reducing troubleshooting time.

TEACHING EXPERIENCE

- **Teaching Assistant, CS501: Mathematical Foundations of Data Science, Rutgers University** 2026 Spring
- **Teaching Assistant, CS462: Introduction to Deep Learning, Rutgers University** 2025 Fall
- **Teaching Assistant, CS520: Introduction to AI, Rutgers University** 2025 Fall

SKILLS

- **Computer Science:** Programming Languages: Python, C++, Java; *ML & DL:* PyTorch, TensorFlow, LoRA/PEFT, RAG; *Data:* Elasticsearch, Scrapy, Pandas, Docker
- **Languages:** Mandarin (Native), English (Proficient)